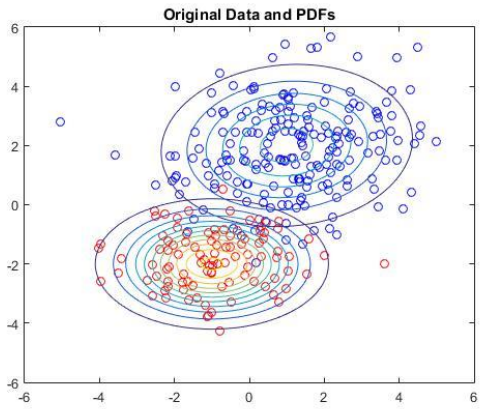
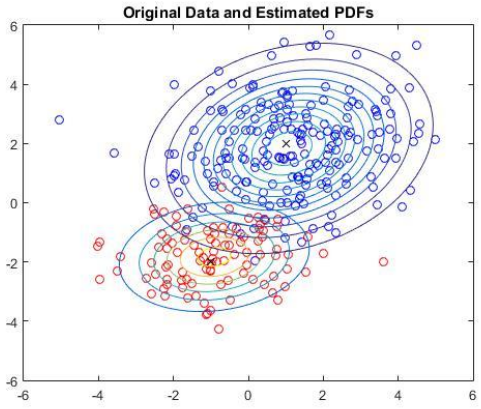
4105109 Computer Vision

Project #2 – Gaussian Mixture Model

Deadline : 12/02(Thu) 11:59 p.m.





1. Trace the matlab code of Gaussian Mixture Model. Write your report:
   - Part 1: Trace the code to understand how EM algorithm works for GMM (GMM_Examples: GMMExample_2D.m). Get to know each parameter in this method and each variable in this code. Include the source code in your report. Write detailed program comments to explain **each line** (what it calculates? how it calculates?). Submit the GMMExample_2D.m which contains your program comments.
   - Part 2: Write a report to explain Gaussian Mixture Model and how EM algorithm solves its parameters in the program flow.
2. What are all parameters that estimate the final PDFs? Adjust these parameters by yourself to see if you get different results?

Note:

- Hand in the matlab code and the report to E-Course.
- Assignment format
  - Zip all your files into a single one and upload it to the E-Course website.
  - Please format the file name as:
    Student ID_proj2_verNo
    ex：602410143_proj2_v1